

Supply Chain Management

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Ch 11. Managing Economies of Scale in a Supply Chain: Cycle Inventory



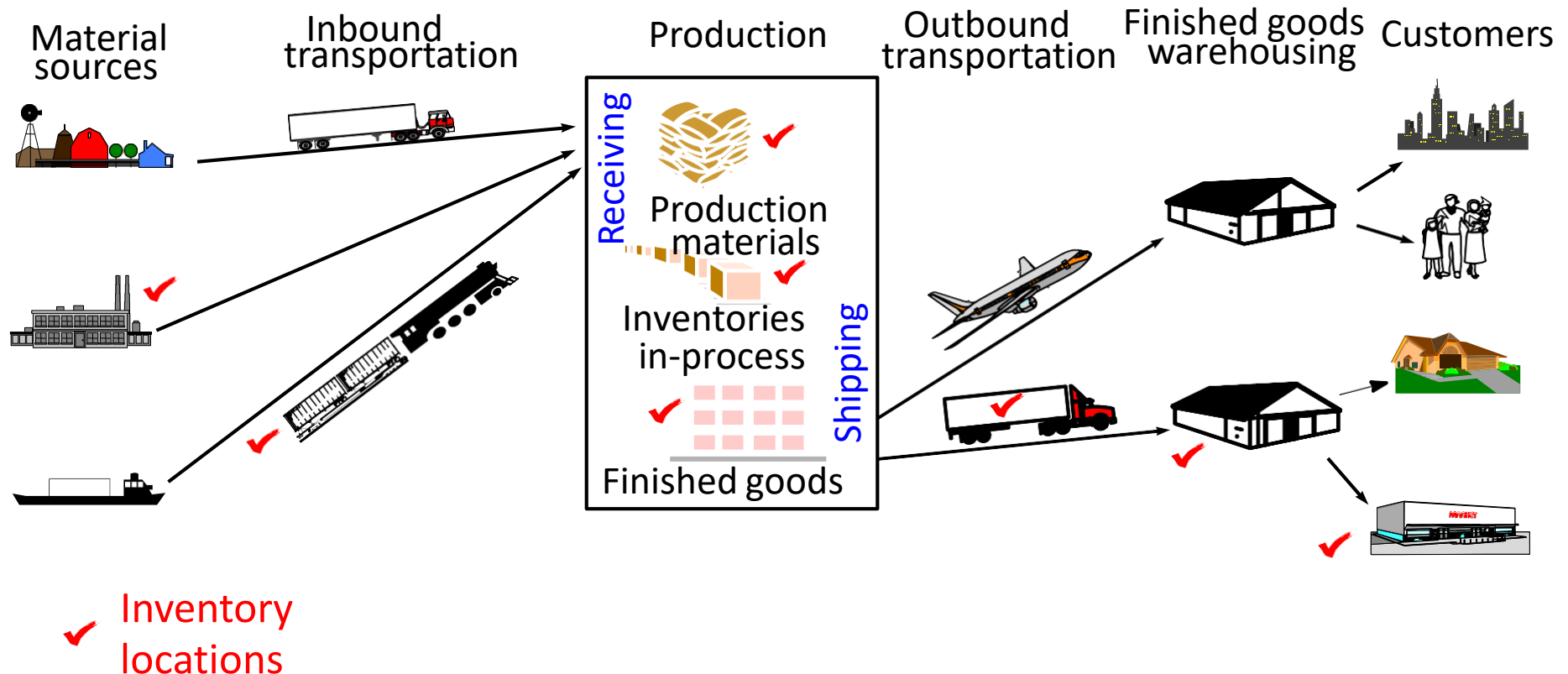
Chapter Overview

1. The Role of Cycle Inventory in a Supply Chain
2. Economies of Scale to Exploit Fixed Costs
3. Aggregating Multiple Products in a Single Order
4. Economies of Scale to Exploit Quantity Discounts
5. Why do Suppliers offer Quantity Discounts?
6. Short-Term Discounting: Trade Promotions
7. Managing Multi-echelon Cycle Inventory
8. Managerial Levers to Reduce Cycle Inventory



11. Managing Economies of Scale in a Supply Chain: Cycle Inventory

- Where are inventories?



11. Managing Economies of Scale in a Supply Chain: Cycle Inventory

- Reasons for Inventories

- To improve customer service (or product availability)
- To encourage production, purchase, and transportation economies
- To act as a hedge against price changes
- To protect against uncertainties in demand and lead times
- To act as a hedge against contingencies (or risk) such as strikes, fire, and disruptions in supply



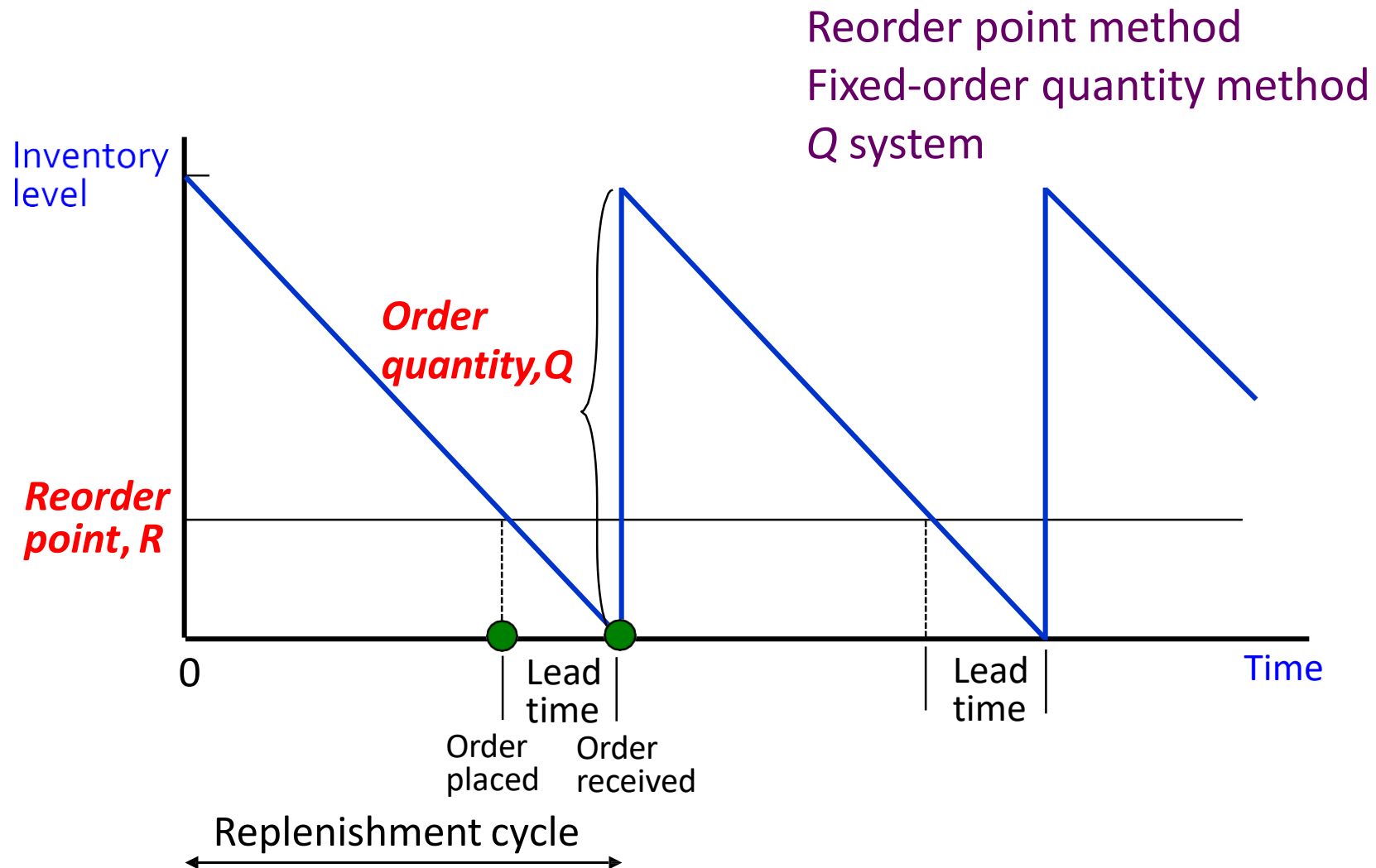
11. Managing Economies of Scale in a Supply Chain: Cycle Inventory

- Reasons Against Inventories
 - Consume capital resources (that might be put to better use elsewhere in the firm)
 - Too often mask quality problems (that would more immediately be solved)
 - Divert management's attention away from careful planning and control as well as collaboration between stages in the supply and distribution channels, by promoting an insular attitude about channel management



11. Managing Economies of Scale in a Supply Chain: Cycle Inventory

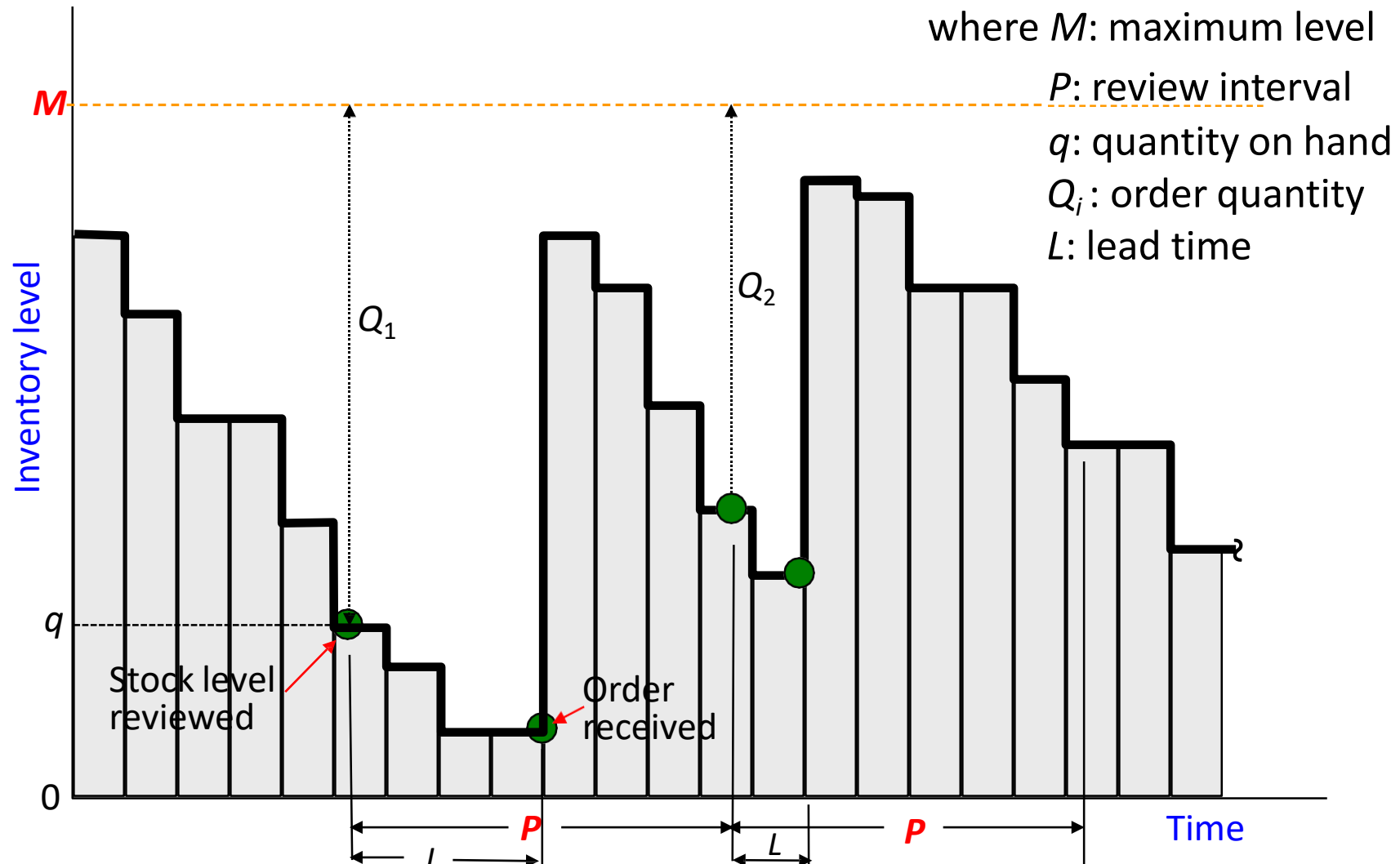
- Continuous Review Method



11. Managing Economies of Scale in a Supply Chain: Cycle Inventory

- Periodic Review Method

Fixed-period method, P system



Lecture Material for Supply Chain Management, by B. D. Song.

Supply Chain Management: Strategy, Planning, and Operation (Sunil Chopra, Pearson, 7th edition)



11. Managing Economies of Scale in a Supply Chain: Cycle Inventory

- Continuous vs. Periodic Review Methods

Continuous review method (Q) is good for products

1. of high value
2. of fast moving

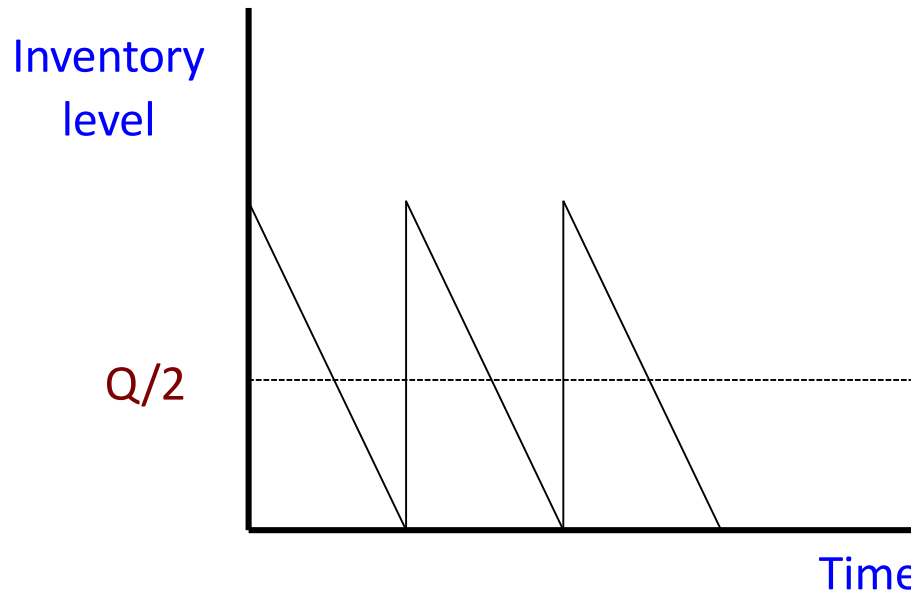
Periodic review method (P) is good for products

1. of low value
2. of slow moving
3. having economies of scale in production, purchasing, or transportation by aggregation of products



11.1. The Role of Cycle Inventory in a Supply Chain

- Cycle inventory: Average inventory that builds up in the supply chain due to the productions or purchases in lots
 - Cycle inventory = $Q/2$ (depends directly on lot size, Q)
- Average flow time resulting from cycle inventory
Cycle inventory/Demand = $Q/(2D)$



11.1. The Role of Cycle Inventory in a Supply Chain

- Example: Jean-Mart (Sales of Jeans)
 - $Q = 1,000$ units, $D = 100$ units/day
 - Cycle inventory: $Q/2 = 1,000/2 = 500$ units
 - Average flow time resulting from cycle inventory: $Q/2D = 1,000/(2)(100) = 5$ days
 - Cycle inventory adds 5 days to the average amount of time a unit spends in the supply chain
- Lower cycle inventory without increasing costs is better because:
 - Vulnerability to demand changes in the market place is lowered
 - Working capital requirement is lowered
 - Inventory holding cost is lowered



11.1. The Role of Cycle Inventory in a Supply Chain

- Cycle inventory is held primarily to take advantage of economies of scale (i.e., lot) in the supply chain
- Costs that must be considered in lot-sizing decision (when stockouts are not allowed)
 - Fixed ordering cost = S (\$/lot)
 - Price paid per unit purchased = C (\$/unit)
 - Holding cost = $H = h \cdot C$ (\$/unit/year)
(where h is a fraction of the unit cost of the product)
- ***The optimal lot size*** is the quantity that minimizes the sum of ordering, material, and holding costs



11.1. The Role of Cycle Inventory in a Supply Chain

- Ideally, cycle inventory decisions should be made considering the total cost across the entire supply chain, however, in practice, each stage generally makes its cycle inventory decisions independently
- Any stage of the supply chain exploits economies of scale in its replenishment decisions in the following three typical situations:
 - 1) A fixed cost is incurred each time an order is placed or produced
 - 2) The supplier offers price discounts based on the quantity purchased per lot
 - 3) The supplier offers short-term price discounts or holds trade promotions



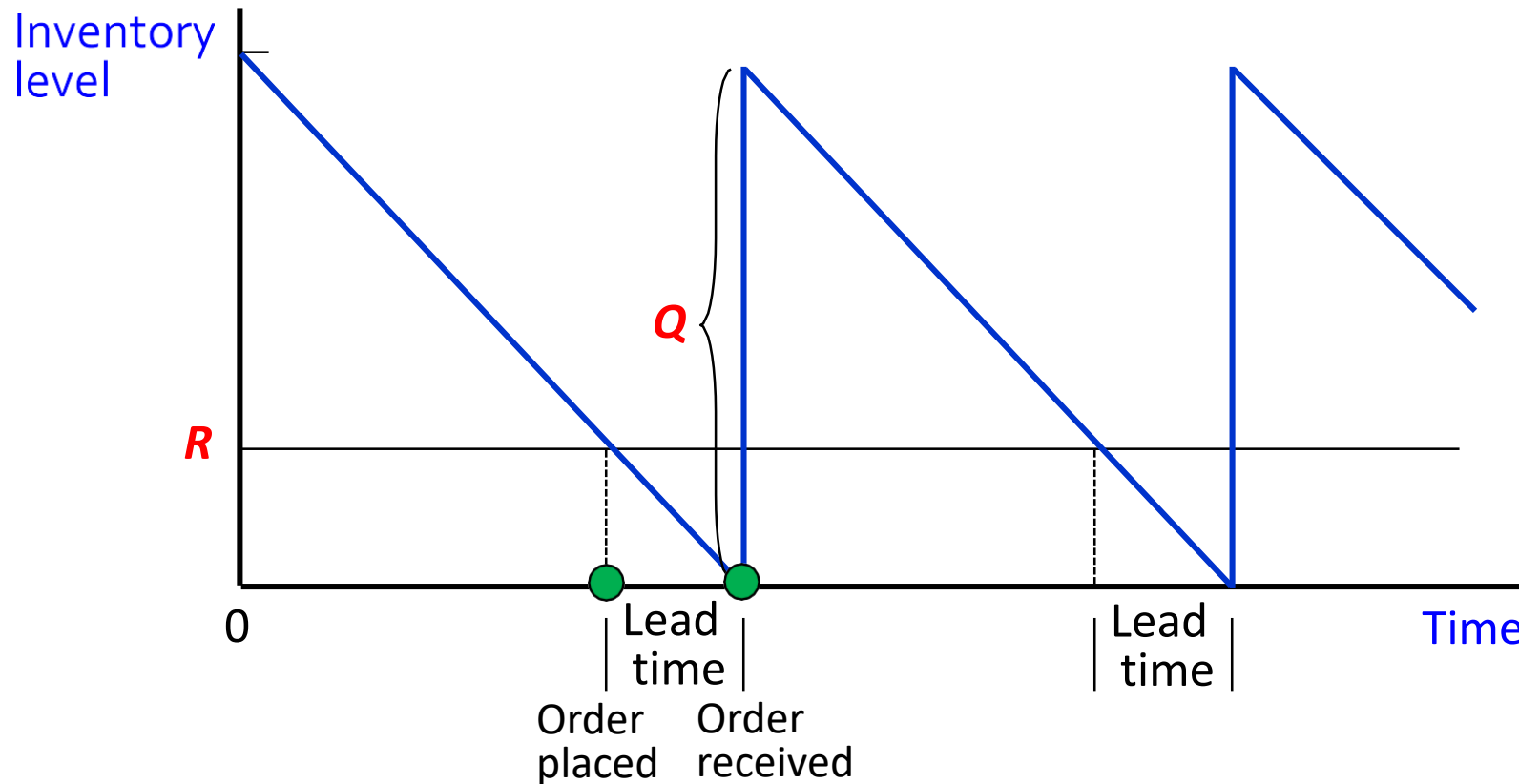
11.2. Economies of Scale to Exploit Fixed Costs

- Estimating cycle inventory related costs in practice
 - Inventory holding cost
 - Costs of capital
 - Obsolescence (or spoilage) cost
 - Handling costs
 - Occupancy (space) costs
 - Miscellaneous costs: Theft, security, damage, tax, insurance
 - Ordering cost
 - Buyer time
 - Transportation costs
 - Receiving costs (works for unloading, inspection, record update)
 - Setup cost for production
 - Costs for loading, unloading, tooling, machine setting, etc.



11.2. Economies of Scale to Exploit Fixed Costs

- Optimal lot sizing for a single product



11.2. Economies of Scale to Exploit Fixed Costs

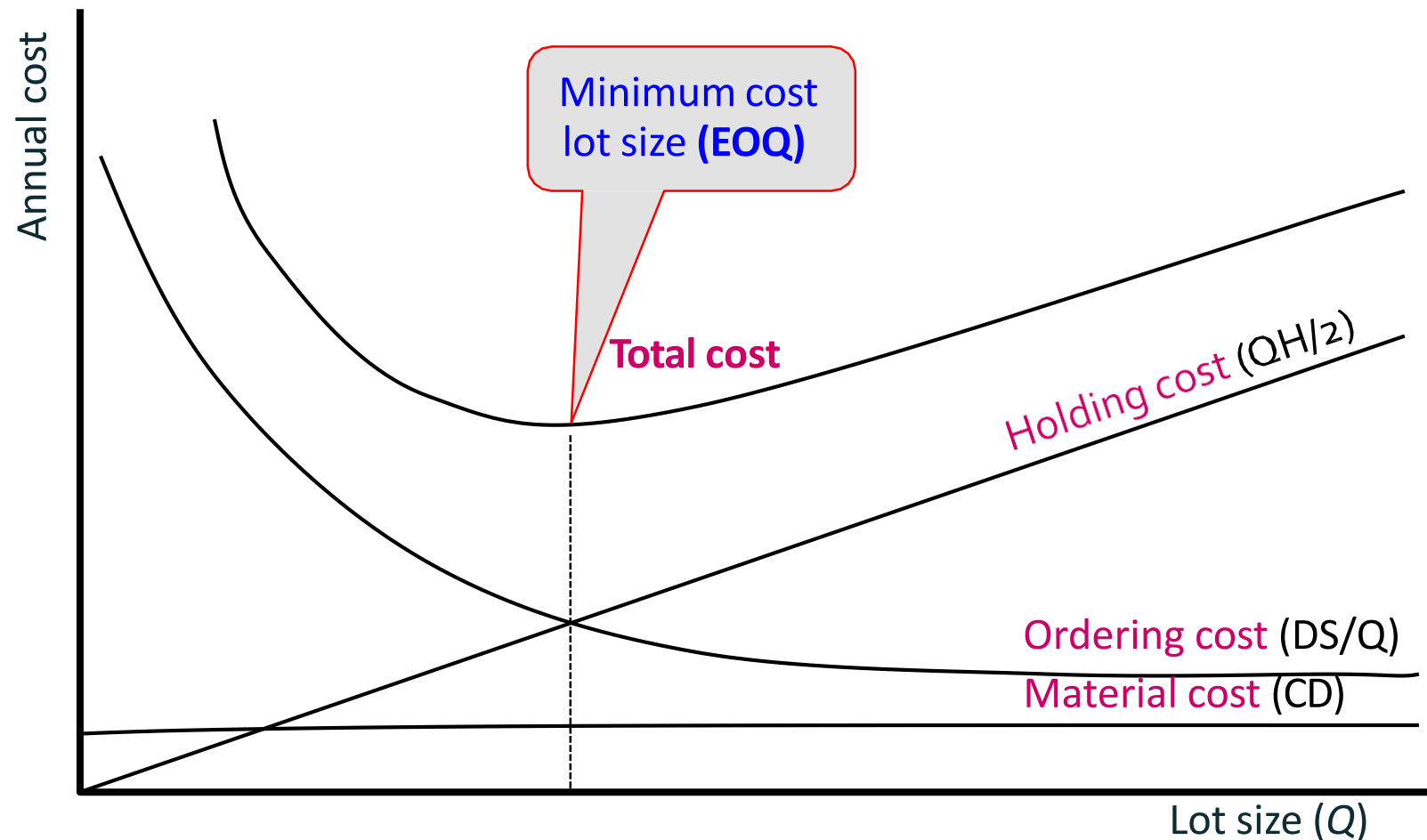
- Lot sizing for a single product
 - Annual demand = D
 - Number of orders per year = D/Q
 - Annual material cost = CD
 - Annual ordering cost = $(D/Q)S$
 - Annual holding cost = $(Q/2)H$
- Assume that the material cost is constant, then total annual cost is:
 $TC = CD + (D/Q)S + (Q/2)hC$

$$\begin{aligned} H &= hC \\ Q^* &= \sqrt{\frac{2DS}{H}} \\ n^* &= \frac{D}{Q} = \sqrt{\frac{DH}{2S}} = \sqrt{\frac{DhC}{2S}} \end{aligned}$$



11.2. Economies of Scale to Exploit Fixed Costs

- Economic Order Quantity



11.2. Economies of Scale to Exploit Fixed Costs

- Example 11.1: Economic Order Quantity - Best Buy: Deskpro Computer
 - $D = 12,000$ units/year, $C = \$500$, $h = 0.2$, $S = \$4,000$ /order
 - Analysis



11.2. Economies of Scale to Exploit Fixed Costs

- Example 11.1: Economic Order Quantity - Best Buy: Deskpro Computer
 - Suppose lot size is reduced to $Q = 200$ units (from 980 units)
 - Flow time:

$$\frac{200}{2(1,000)} = 0.1 \text{ months } (< 0.49 \text{ months})$$

- Annual ordering and holding cost

$$(12,000/200)(4,000) + (200/2)(0.2)(500) = \$250,000 (> \$97,980)$$

To make it economically feasible to reduce lot size, the fixed cost associated with each lot would have to be reduced



11.2. Economies of Scale to Exploit Fixed Costs

- Example 11.2: Relationship between lot size and ordering cost
 - If desired lot size, $Q^* = 200$ units, what would S have to be?
(When $D = 12,000$ units, $C = \$500$, $h = 0.2$, $S = \$4,000/\text{order}$, $Q^* = 980$)
 - Analysis: Use EOQ equation and solve for S

$$S = \frac{hCQ^{*2}}{2D} = [(0.2)(500)(200)^2]/[(2)(12,000)] = \$166.7 (< \$4,000)$$

To reduce optimal lot size by a factor of k , the fixed ordering cost must be reduced by a factor of k^2



11.2. Economies of Scale to Exploit Fixed Costs

- Key Points from EOQ model
 - In deciding the optimal lot size, the trade-off is between fixed ordering (or setup) cost and holding cost
 - **If demand increases by a factor of k** , it is optimal to increase lot size by a factor of $k^{1/2}$. In this case, the number of orders placed per year increase by a factor of $k^{1/2}$ and flow time resulting from cycle inventory decrease by a factor of $k^{1/2}$
 - Proof for flow time:

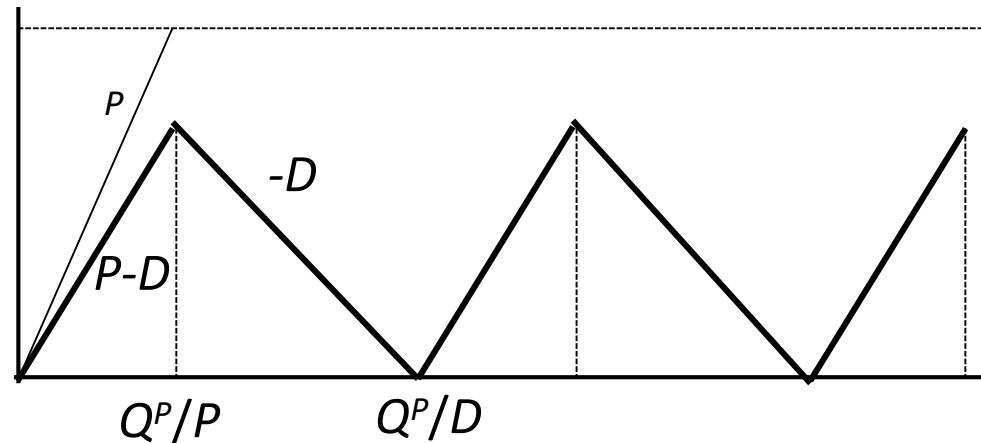
$$\frac{Q/2}{D} \rightarrow \left(\frac{\sqrt{k}}{k} = \frac{1}{\sqrt{k}} \right)$$

- **If lot size is to be reduced by a factor of k** , fixed ordering cost should be reduced by a factor of k^2



11.2. Economies of Scale to Exploit Fixed Costs

- Production Lot Sizing
 - Economic production quantity (EPQ) model



$$TC(Q^P) = \frac{DS}{Q^P} + \frac{(P-D) \frac{Q^P}{P}}{2} hC$$

$$Q^P = \sqrt{\frac{2DS}{(1-D/P)hC}} = EOQ \sqrt{\frac{P}{P-D}}$$

where

Q^P : EPQ

P : production rate ($P > D$)

D : demand rate



11.3. Aggregating Multiple Products in a Single Order

- Aggregating replenishment across products in a single order allows for a **reduction in lot size for individual products**, because transportation and other fixed ordering costs are spread across multiple products
- Other ways to achieve aggregation include having a single delivery coming from multiple suppliers or having a single delivery to multiple retailers
(e.g.) Wal-Mart's or Seven-Eleven Japan's cross-docking at DC
- As more product are included in a single order when attempting to reduce lot sizes, try to reduce the receiving and storing costs which are fixed costs
(e.g.) Advanced shipping notices (ASN), RFID technique



11.3. Aggregating Multiple Products in a Single Order

- Example: Best Buy: Four Computer Models (Data from Example 11.1)
 - Assume demand for each model is 1,000 units per month Fixed transportation cost = \$4,000, $h \cdot C = 0.2(\$500) = \$100/\text{year}$
 - **If each model is ordered separately:**
 - $Q^* = 980$ units for each model
 - Total cycle inventory = $4(Q/2) = (4)(980)/2 = 1,960$ units
 - **Aggregating four models in a single order:**

$$\text{Combined } Q^* = \sqrt{\frac{2(4)(12)(1000)(4000)}{100}} \cong 1,960 \text{ units}$$

- For each model: $Q^* = 1,960/4 = 490$ units
- Total cycle inventory = $(4)(490/2) = 980$ units ↓
- Average flow time and inventory holding costs will be reduced



11.3. Aggregating Multiple Products in a Single Order

- Lot sizing with multiple products
 - For the fixed ordering cost,
 - **Common or joint order cost (S)**: Independent of product variety on the truck
(e.g.) Transportation cost
 - **Product-specific order cost (s_i)**: Dependent on product and increases with variety on the truck
(e.g.) Loading cost, receiving cost, storing cost
 - Best Buy considers three approaches to the lot sizing decision:
 - Lots are ordered and delivered independently for each product model
 - Lots are ordered and delivered jointly for all models
 - Lots are ordered and delivered jointly for a selected subset of models
(i.e., tailored aggregation)



11.3. Aggregating Multiple Products in a Single Order

- Example (11.3, 11.4, 11.6): Lot sizing with multiple products
 - Best Buy: Three Computer Models (Litepro, Medpro, Heavypro)
 - Demand for three computer models per year:
 $D_L = 12,000$; $D_M = 1,200$; $D_H = 120$
 - Common transportation cost:
 $S = \$4,000$
 - Product-specific order cost for receiving and storage:
 $s_L = \$1,000$; $s_M = \$1,000$; $s_H = \$1,000$
 - Annual holding cost: $h = 0.2$
 - Unit cost: $C_L = \$500$; $C_M = \$500$; $C_H = \$500$



11.3. Aggregating Multiple Products in a Single Order

- Example 11.3: Order each product independently

	Litepro	Medpro	Heavypro
Demand per year	12,000	1,200	120
Fixed cost/order	\$5,000	\$5,000	\$5,000
Optimal order size	1,095	346	110
Cycle inventory	548	173	55
Annual holding cost	\$54,772	\$17,321	\$5,477
Order frequency	11.0 year	3.5 year	1.1 year
Annual ordering cost	\$54,772	\$17,321	\$5,477
Average flow time	2.4 weeks	7.5 weeks	23.7 weeks
Annual cost	\$109,544	\$34,642	\$10,954

– Total annual order and holding cost = \$155,140

(note)

$$EOQ_{Litep} = \sqrt{\frac{2(12,000)(4000+1000)}{0.2(500)}} = 1,095$$



11.3. Aggregating Multiple Products in a Single Order

- Example 11.4: Products ordered and delivered jointly (Aggregation)
 - $S^* = S + s_L + s_M + s_H = \$4,000 + \$1,000 + \$1,000 + \$1,000 = \$7,000$ per order
 - Using **Eq. 11.7** in p.291:

$$n^* = \frac{D}{Q} = \sqrt{\frac{\sum_{i=1}^k D_i h C_i}{2S^*}} = 9.75 \text{ orders}$$
$$Q_L = D_L/n^* = 12,000/9.75 = 1,230 \text{ units}$$
$$Q_M = D_M/n^* = 1,200/9.75 = 123 \text{ units}$$
$$Q_H = D_H/n^* = 120/9.75 = 12.3 \text{ units}$$

- Total annual order cost = 9.75 times $\times$ \$7,000 = \$68,250
- Total annual order and holding cost = \$136,528 (13% decrease)



11.3. Aggregating Multiple Products in a Single Order

- Lots are ordered and delivered jointly for a selected subset of models
 - Tailored aggregation
 - Five-step procedure (for near optimal solution, p.294)
 - **Example 11.6**
 - Solution: (L, M, H), (L), (L,M), (L), (L,M), (L), (L, M, H), (L),
 - Total annual cost = \$130,767 (**4% reduction**) < \$136,528



11.3. Aggregating Multiple Products in a Single Order

- Example: Lot sizes and costs for ordering policy using heuristic

	Litepro	Medpro	Heavypro
Demand per year (D)	12,000	1,200	120
Order frequency (n^*)	11.47 year	5.74 year	2.29 year
Optimal order size (D/n^*)	1,046	209	52
Cycle inventory	523	104.5	26
Annual holding cost	\$52,307	\$10,461	\$2,615
Average flow time	2.27 weeks	4.53 weeks	11.35 weeks



11.3. Aggregating Multiple Products in a Single Order

- Lessons from aggregation
 - Aggregation across products, customers (retailers), or suppliers allows firms to lower lot size (or cycle inventory) without increasing cost
 - **Complete aggregation** is effective if product-specific fixed cost is a small fraction of common fixed cost and demand difference among products are not significant
 - **Tailored aggregation** is effective if product-specific fixed cost is a large fraction of common fixed cost and demand difference among products are significant



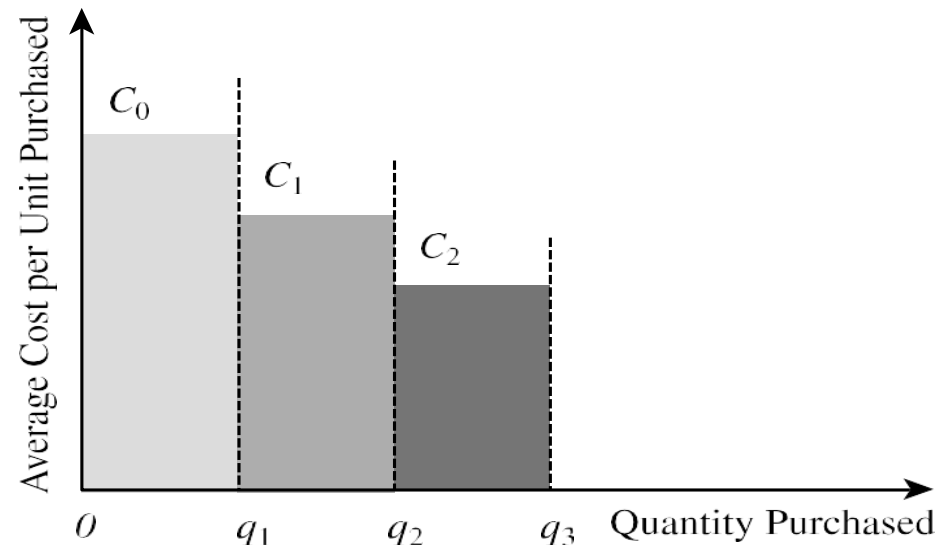
11.4. Economies of Scale to Exploit Quantity Discounts

- Quantity discount
 - Lot size-based quantity discounts
 - All unit quantity discount
 - Marginal unit quantity discount
 - Volume-based quantity discounts
- Two basic questions
 - Given a discount-pricing schedule, what is the optimal purchasing decision **for a buyer** seeking to maximize profits?
 - Under what conditions should a supplier offer quantity discounts? What are appropriate pricing schedules **for a supplier (or manufacturer)** seeking to maximize profits?
- Quantity discounts are valuable in a supply chain because they improve coordination to increase total supply chain profits



11.4. Economies of Scale to Exploit Quantity Discounts

- All-unit quantity discounts
 - Pricing schedule has specified quantity break points $q_0, q_1, \dots, q_r$, where $q_0 = 0$
 - If an order is placed that is at least as large as q_i but smaller than q_{i+1} , then each unit is obtained at a cost of C_i
 - The objective for the buyer (“a retailer” in our example) is to decide on a lot size that will maximize profits, or equivalently, minimize the sum of material, order, and holding costs



11.4. Economies of Scale to Exploit Quantity Discounts

- All-unit quantity discounts: EOQ and total cost
 - Suppose there are three possible cases for order quantity Q
 - If $Q < q_1$, the unit price is C_0
 - If $q_1 \leq Q < q_2$, the unit price is C_1
 - If $q_2 \leq Q$, the unit price is C_2 (where $C_2 < C_1 < C_0$)
 - Economic order quantity (EOQ) for price C_i

$$Q_i = \sqrt{\frac{2DS}{hC_i}}$$

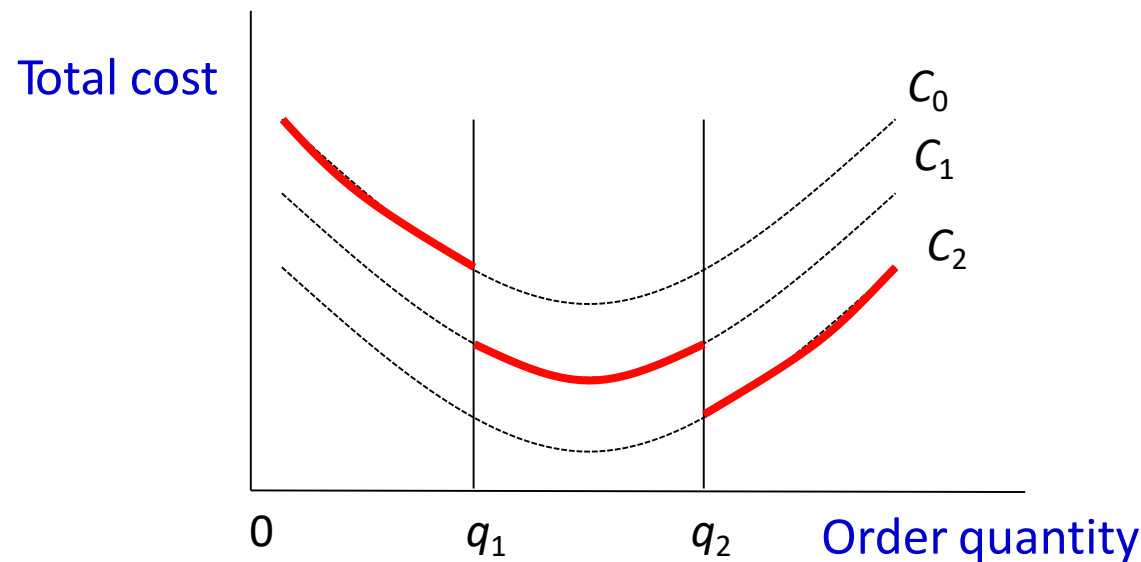
- Total annual cost of ordering Q_i units for price C_i

$$TC_i = \left(\frac{D}{Q_i}\right)S + \frac{Q_i}{2}hC_i + DC_i$$



11.4. Economies of Scale to Exploit Quantity Discounts

- All-unit quantity discounts: Total cost curves
 - Determination of the EOQ under all-unit quantity discount
 - Find the optimal order quantity for each unit price C_i ($C_2 < C_1 < C_0$):
 - If $Q_2 < q_2$ for C_2 , the ordering cost of q_2 is minimal
 - If $q_1 \leq Q_1 < q_2$ for C_1 , the ordering cost of EOQ is minimal
 - If $Q_0 > q_1$ for C_0 , the ordering cost of q_1-1 is minimal
 - Choose the quantity whose the total cost is minimal among them



11.4. Economies of Scale to Exploit Quantity Discounts

- All-unit quantity discounts procedure

(Different from what is in pp.297-298 of the textbook)

Step 1: Calculate the EOQ for the lowest price. If it is feasible, stop. This is the optimal lot size. Calculate the total cost (TC) for that lot size.

Step 2: If the EOQ is not feasible, calculate the TC for the smallest feasible quantity for that price.

Step 3: Calculate the EOQ for the next lowest price. If it is feasible, stop and calculate the TC for that quantity and price.

Step 4: Compare the TC for *Steps* 2 and 3 and choose the quantity corresponding to the lowest TC .

Step 5: If the EOQ in Step 3 is not feasible, repeat *Steps* 2, 3, and 4 until a feasible EOQ is found.



11.4. Economies of Scale to Exploit Quantity Discounts

- Example 11.7: All-unit quantity discounts
 - Drugs Online (DO): On-Line Vitamin Sale Retailer

<u>Order quantity</u>	<u>Unit price</u>
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0~4,999	\$3.00
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5000~9,999	\$2.96
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10,000 or more	\$2.92
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$$q_0 = 0, q_1 = 5,000, q_2 = 10,000$$

$$C_0 = \$3, C_1 = \$2.96, C_2 = \$2.92$$

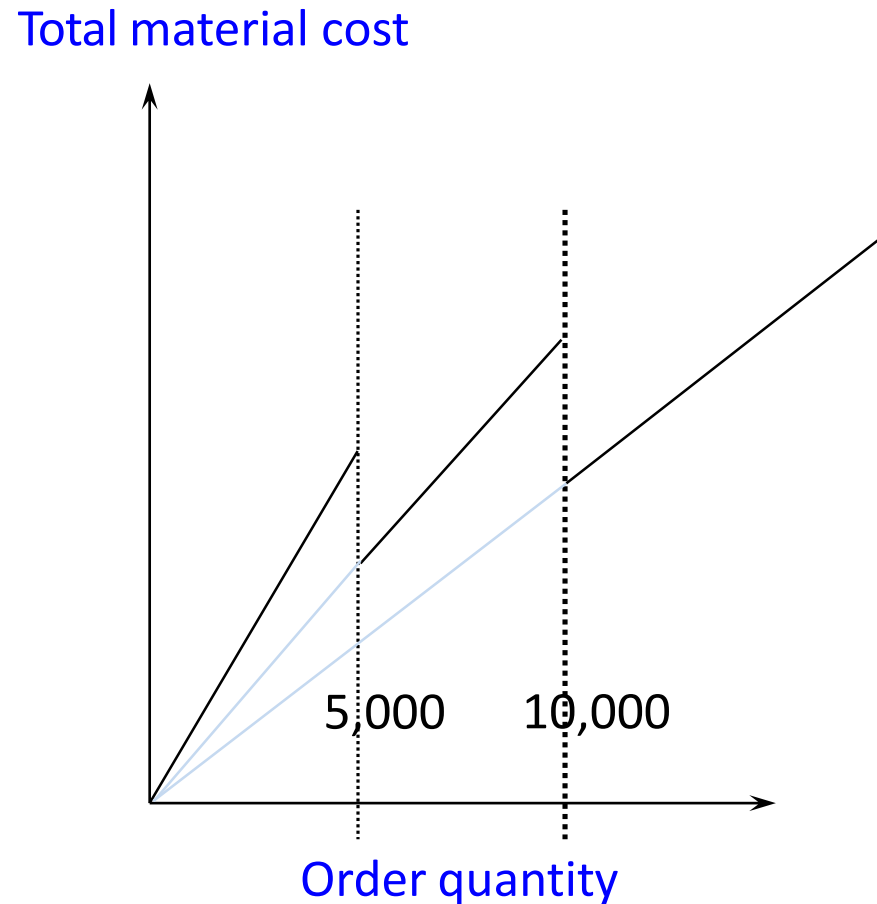
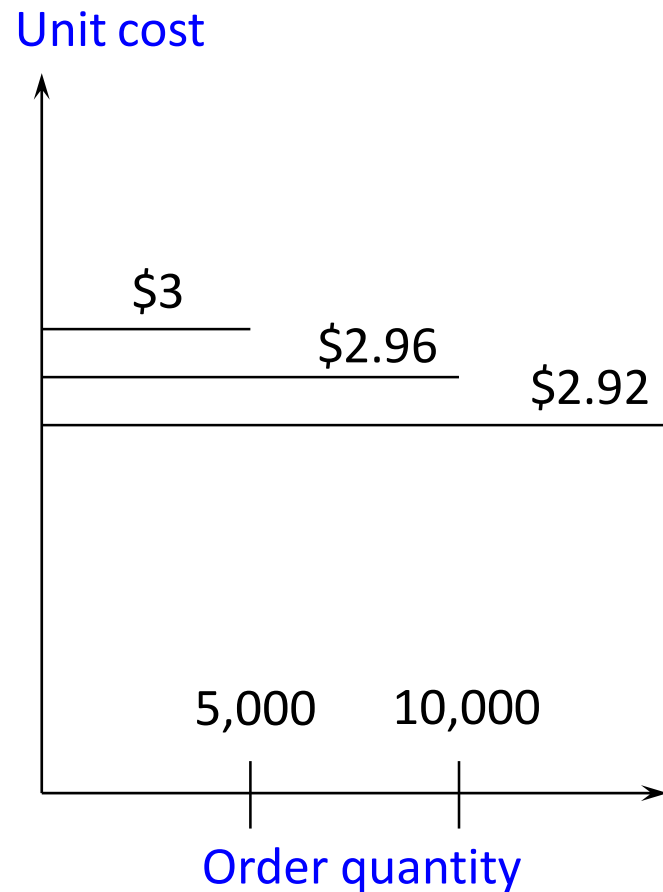
$$D = 120,000 \text{ bottles/year}, S = \$100 \text{ per order}, h = 0.2$$

Evaluate number of bottles that DO manager should order in each lot?



11.4. Economies of Scale to Exploit Quantity Discounts

- Example 11.7: All-unit quantity discounts
 - Drugs Online (DO): On-Line Vitamin Sale Retailer



11.4. Economies of Scale to Exploit Quantity Discounts

- Example 11.7: All-unit quantity discounts

Step 1: Calculate $Q_2^* = \sqrt{(2DS)/hC_2}$

$$= \sqrt{(2)(120,000)(100)/(0.2)(2.92)} = 6,410 \text{ bottles}$$

- Not feasible ($6,410 < 10,000$)
- Thus, calculate TC_2 using $C_2 = \$2.92$ and $q_2 = 10,000$ bottles
- $TC_2 = (120,000/10,000)(100) + (10,000/2)(0.2)(2.92) + (120,000)(2.92)$
= \$354,520

Step 2: Calculate $Q_1^* = \sqrt{(2DS)/hC_1}$

$$= \sqrt{[(2)(120,000)(100)/(0.2)(2.96)]} = 6,367 \text{ bottles}$$

Feasible ($5,000 < 6,367 \leq 10,000$) → **Stop!**

$$TC_1 = (120,000/6,367)(100) + (6,367/2)(0.2)(2.96) + (120,000)(2.96)$$

= \$358,969

$TC_2 < TC_1 \rightarrow$ The optimal order quantity, $Q^* = 10,000$ bottles



11.4. Economies of Scale to Exploit Quantity Discounts

- All-unit quantity discounts: Summary
 - Suppose the manufacturer sold all bottles for \$3 in Example 11.7
 - Q^* for DO would be 6,324 bottles (< 10,000 bottles with discount)
 - What is the effect of such a discount schedule?
 - Retailers are encouraged to increase their order size
 - Average inventory (i.e., cycle inventory) in the supply chain is increased
 - Average flow time is increased
 - **Is an all-unit quantity discount an advantage in the supply chain?**



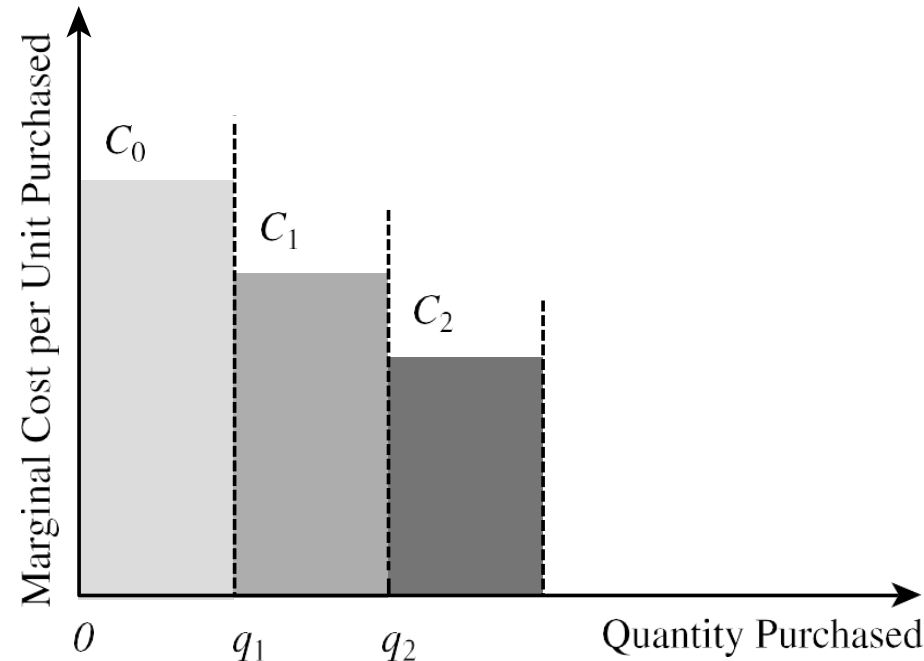
11.4. Economies of Scale to Exploit Quantity Discounts

- Quantity discount
 - Lot size-based quantity discount
 - All-unit quantity discount
 - Marginal unit quantity discount
 - Volume-based quantity discount



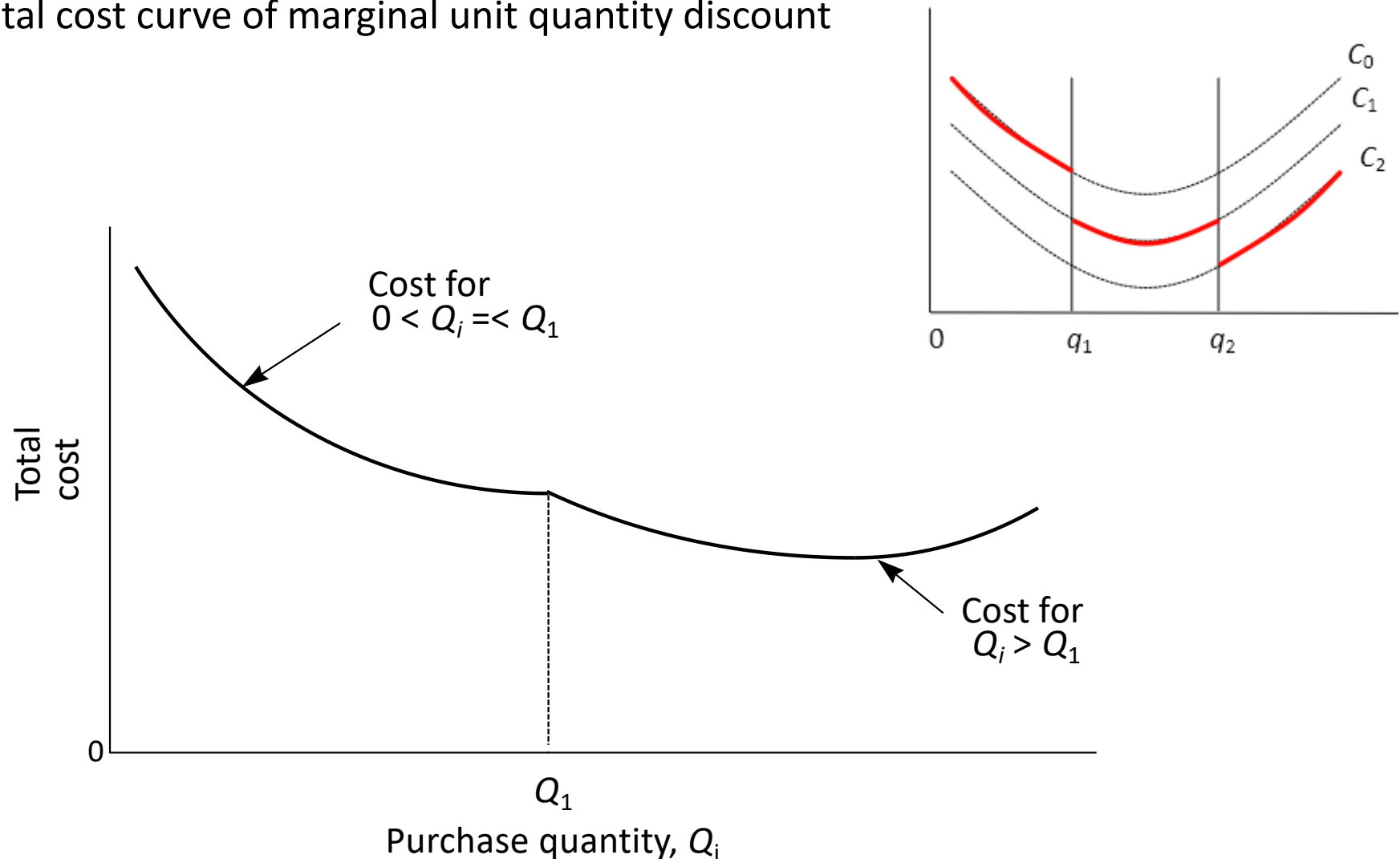
11.4. Economies of Scale to Exploit Quantity Discounts

- Marginal unit quantity discount
 - Marginal unit quantity discount
 - Price discount only applies to the items within the price break quantity
 - Average price continues to drop with increasing purchase quantities



11.4. Economies of Scale to Exploit Quantity Discounts

- Marginal unit quantity discount
 - Total cost curve of marginal unit quantity discount



11.4. Economies of Scale to Exploit Quantity Discounts

- Marginal unit quantity discount
 - Example 11.8 (substitute for Example 11.8 in the textbook)
 - First 500 units: $C = \$5$ per unit
 - All units over 500 units: $0.95(\$5) = \4.75 per unit
 - $D = 2,600$ units per year, $h = 0.1$ per year, $S = \$10$ per order



11.4. Economies of Scale to Exploit Quantity Discounts

- Marginal unit quantity discount

- Example 11.8 (substitute for Example 11.8 in the textbook)

<i>Q</i>	<i>Average unit price (C)</i>	<i>CD</i>	<i>DS/Q</i>	<i>hCQ/2</i>	<i>Annual total cost</i>
300	\$5	\$13,000	\$86.67	\$75.00	\$13,162
400	5	13,000	65.00	100.00	13,165
500	5	13,000	52.00	125.00	13,177
600	$(500 \times 5 + 100 \times 4.75) / 600 = 4.96$	12,896	43.33	148.80	13,088
800	$(500 \times 5 + 300 \times 4.75) / 800 = 4.91$	12,756	32.50	196.25	12,985
900	$(500 \times 5 + 400 \times 4.75) / 900 = 4.89$	12,711	28.89	220.00	12,960
1,000	$(500 \times 5 + 500 \times 4.75) / 1000 = 4.88$	12,675	26.00	243.75	12,945
1,500	$(500 \times 5 + 1000 \times 4.75) / 1500 = 4.86$	12,567	17.33	362.50	12,947
2,000	$(500 \times 5 + 1500 \times 4.75) / 2000$	12,513	13.00	481.25	13,007



11.5. Why Do Suppliers Offer Quantity Discount?

- To increase the supplier profit (so, total supply chain profit) through coordination between the supplier (or manufacturer) and retailer under some circumstances
 - Case 1: For commodity products
 - A competitive market exists (e.g.) Milk, vitamin
 - Market sets the price
 - Firm's objective is to lower costs
 - Case 2: For products for which the firm has market power
 - Retailers set prices (e.g.) New products
 - Demand is influenced by the price
(e.g.) $D = 360,000 - 60,000p$
 - Firm's objective is to higher the profit



11.5. Why Do Suppliers Offer Quantity Discount?

- Example 11.9: The impact of locally optimal lot sizes on a supply chain (Case 1)
 - DO: On-Line Vitamin Sale Retailer
 - Each time DO places an order, the manufacturer pack the bottles at a steady rate that matches demand
 - Demand for vitamins at DO = 10,000 bottles/month
 - $S_R = \$100$ per order, $h_R = 0.2$, $C_R = \$3/\text{bottle}$ (Retailer)
 - $S_M = \$250$ per order, $h_M = 0.2$, $C_M = \$2/\text{bottle}$ (Manufacturer)
 - Analysis:
 - Optimal order size for DO: $Q_R = 6,325$ bottles $= \sqrt{\frac{2(120,000)(100)}{0.2(3)}}$
 - Annual cost for DO: $(D/Q_R)S_R + Q_R/2 h_R C_R = \$3,795$
 - Annual cost for manufacturer: $(D/Q_R)S_M + (Q_R/2) h_M C_M = \$6,008$
 - Thus, total supply chain cost = \$9,803



11.5. Why Do Suppliers Offer Quantity Discount?

- Example 11.9: The optimal lot size from the supply chain prospective (Case 1)
 - The optimal lot size from the supply chain prospective:

$$\begin{array}{l} \text{Annual cost for} \\ \text{DO and manufacturer} \end{array} = \left(\frac{D}{Q}\right)S_R + \left(\frac{Q}{2}\right)h_R C_R + \left(\frac{D}{Q}\right)S_M + \left(\frac{Q}{2}\right)h_M C_M$$

$$Q^* = \sqrt{\frac{2D(S_R + S_M)}{h_R C_R + h_M C_M}} = 9,165 > 6,325 \text{ bottles}$$

$$\text{Annual cost for DO} = \left(\frac{D}{Q^*}\right)S_R + \left(\frac{Q^*}{2}\right)h_R C_R = \$4,059 > \$3,795$$

$$\text{Annual cost for manufacturer} = \left(\frac{D}{Q^*}\right)S_M + \left(\frac{Q^*}{2}\right)h_M C_M = \$5,106 < \$6,008$$

$$\begin{array}{l} \text{Annual supply chain cost} \\ \text{(DO and manufacturer)} \end{array} = \$5,106 + \$4,059 = \$9,165 < \$9,803$$



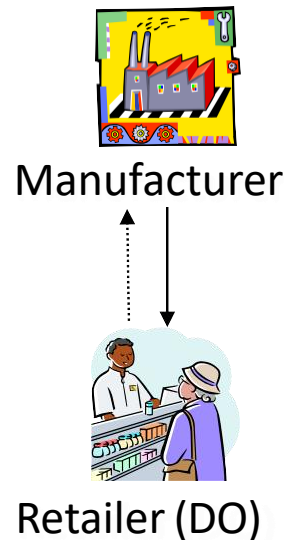
11.5. Why Do Suppliers Offer Quantity Discount?

- Example 11.10: Designing a suitable lot size-based quantity discount
 - What if DO orders in lots of 9,165 instead of 6,325 bottles?
 - Annual supply chain cost = \$9,803 → \$9,165
 - (manufacturer's cost = \$5,106, DO's cost = \$4,059)
 - Manufacturer's cost = \$902/year ↓, DO's cost = \$264/year ↑
 - That is, the supply chain saves **\$638**
 - Effective pricing schemes for a lot size-based quantity discount (to enforce $Q = 9,165$ bottles)
 - All-unit quantity discount for DO
 - \$3 for lots below 9,165
 - \$2.9978 for lots of 9,165 or more ($\$264/120,000 = \0.0022)
 - Manufacturer's profit increase is **\$638** ($= \$902 - \264)
 - The increased lot size increases cycle inventory for DO



11.5. Why Do Suppliers Offer Quantity Discount?

- Coordination for commodity products: Summary
 - Coordination to lower total supply chain cost:



	Q = 6,325 bottles (Retailer's EOQ)	Q = 9,165 bottles (New alternative)	Results
<u>Retailer</u> $S_R = \$100$ $h_R = 0.2$ $C_R = \$3$	Ordering cost: \$1,898 Holding cost: \$1,897 Total cost: \$3,795	Ordering cost: \$1,309 Holding cost: \$2,750 Total cost: \$4,059	\$264 ↑
<u>Manufacturer</u> $S_M = \$250$ $h_M = 0.2$ $C_M = \$2$	Ordering cost: \$4,744 Holding cost: \$1,264 Total cost: \$6,008	Ordering cost: \$3,273 Holding cost: \$1,833 Total cost: \$5,106	\$902 ↓
Total cost	\$9,803	\$9,165	\$638 ↓



11.5. Why Do Suppliers Offer Quantity Discount?

- Quantity discounts when firm has market power (Case 2)
 - What are the optimal prices and profits for a new invented vitamin?

Annual demand equation = $360,000 - 60,000p$ (p : sale price at DO)

$C_M = \$2$ per bottle

- **Analysis:**

(1) Independent pricing

$$\text{Max. } Prof_R = (p - C_R)(360,000 - 60,000p)$$

Using the first derivation, $p = 3 + (C_R/2)$

$$\text{Max. } Prof_M = (C_R - C_M)(360,000 - 60,000p)$$

$$= (C_R - 2)[360,000 - 60,000(3 + C_R/2)]$$

Using the first derivation, $C_R^* = \$4$. Thus, $p^* = \$5$

(2) Coordinated pricing

$$\text{Max. } Prof_{SC} = (p - C_M)(360,000 - 60,000p)$$

Using the first derivation, $p^* = 3 + (C_M/2) = 3 + (2/2) = \4



11.5. Why Do Suppliers Offer Quantity Discount?

- Quantity discounts when firm has market power (Case 2)
 - **Independent pricing**
 - Optimal pricing policy: $p = \$5$ for DO when $C_R = \$4$
 - Total market demand: $360,000 - 60,000p = 60,000$ bottles
 - DO's profit = $(\$5 - \$4) \times 60,000 = \$60,000$
 - Manufacturer's profit = $(\$4 - \$2) \times 60,000 = \$120,000$
 - Supply chain profit = $\$60,000 + \$120,000 = \$180,000$
 - **Coordinated pricing**
 - Optimal pricing policy: $p = \$4$ for DO
 - Total market demand: $360,000 - 60,000p = 120,000$ bottles
 - Supply chain profit = $(\$4 - \$2) \times 120,000 = \$240,000$ (\$60,000↑)



11.5. Why Do Suppliers Offer Quantity Discount?

- Quantity discounts when firm has market power (Case 2)
 - The supply chain profit is lower if each stage of the supply chain makes its pricing decisions independently, with the objective of maximizing its own profit ('Double marginalization')
 - The optimal profit split through pricing schemes must be considered when pricing decisions are made in a coordinated manner



11.5. Why Do Suppliers Offer Quantity Discount?

- Pricing scheme: Example
 - A pricing scheme example that the manufacturer may use to achieve the coordinated solution and maximize supply chain profits (i.e., \$240,000)
 - **Volume-based quantity discount**
 - Retail price (p) = \$4 (then, market demand = 120,000 bottles/year)
 - $C_R = \$4$ for total volume < 120,000 bottles per year
 - $C_R = \$3.5$ for total volume $\geq$ 120,000 bottles per year
 - *Manufacturer*
 - Total profit: $120,000 C_R - \$2 = 120,000(\$3.5 - \$2) = \$180,000$
(\$60,000↑ compared with the independent pricing)
 - *DO*
 - Total profit: $120,000(p - C_R) = 120,000(\$4 - \$3.5) = \$60,000$
(same as for the independent pricing)



11.5. Why Do Suppliers Offer Quantity Discount?

- Lessons from quantity discounts
 - Quantity discounts play a role in supply chain coordination and improve supply chain profits
 - Lot size–based discounts tend to raise the cycle inventory in the supply chain
 - Volume-based discounts are compatible with small lots that reduce cycle inventory in the supply chain
 - Volume-based discounts are better than lot size-based discounts unless the manufacturer has large fixed costs associated with each lot
 - Under volume-based discounts, retailers will tend to increase the size of the lot toward the end of the evaluation period ('hockey stick phenomenon'). Thus, use a 'rolling horizon' in evaluation



11.6. Short-Term Discounting: Trade Promotions

- Trade promotions of manufacturers (or suppliers) are price discounts to retailers for a limited period of time
- A few of key goals for a trade promotions from a manufacturer's perspective:
 - Induce retailers to spur sales to the customer
 - Shift inventory from the manufacturer to the retailer and the customer
 - Smooth production by increasing demand during the off-peak period
 - Defend a brand against competition
- Retailer has two primary options in response to a promotion
 - 1) Pass through some or all of the promotion to customers to spur sales
 - 2) Purchase in greater quantity during promotion period, but pass through very little of savings to customers - Forward buy!



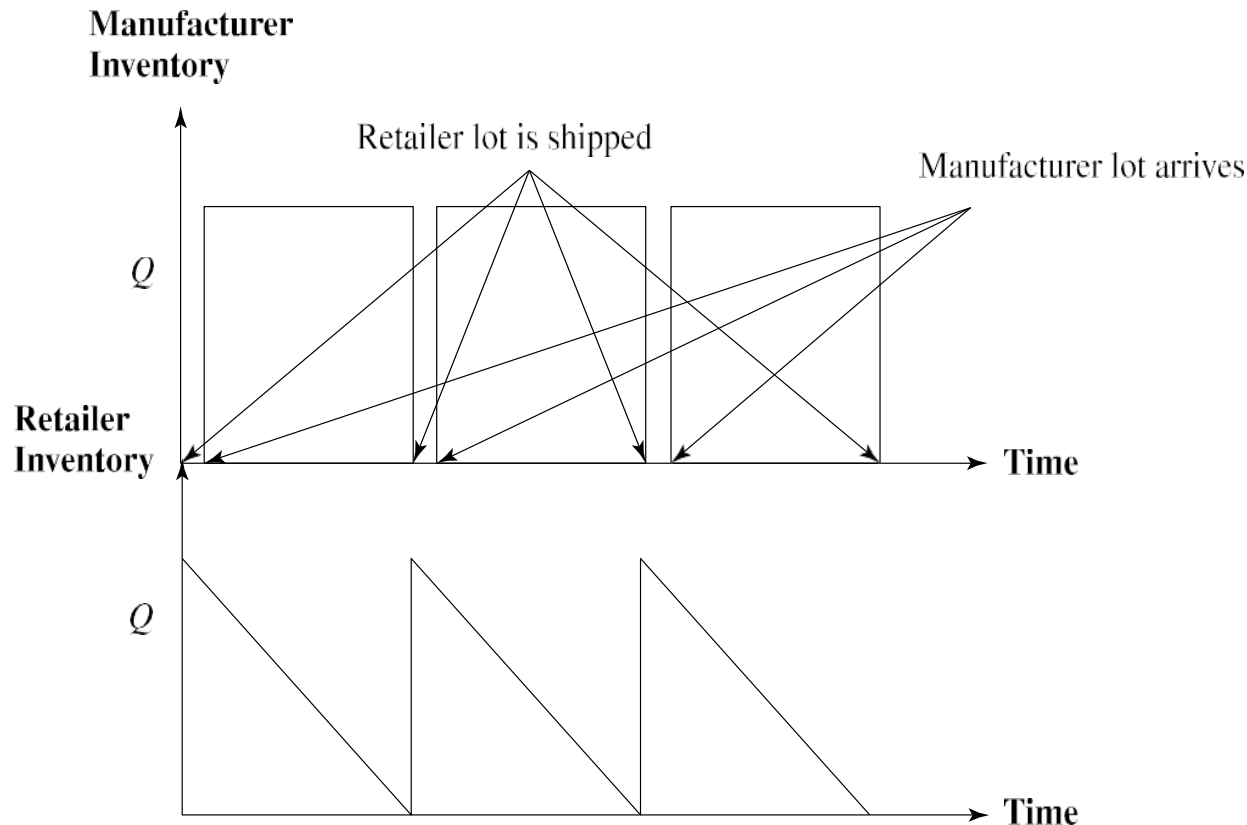
11.6. Short-Term Discounting: Trade Promotions

- When a manufacturer offers a promotion, the goal for the manufacturer is to take actions (countermeasures) **to discourage the retailers' forward buying**
- Countermeasures
 - **EDLP** (every day low pricing): Fixed price over time and no short-term discounts, so no forward buy
 - **Scan-based promotions**: Discounts to the retailers that are based on actual sales to customers (sell-through) rather than the amount purchased by the retailer (sell-in)
 - **Manufacturer coupons**: Discounts directly to the customers using manufacturer coupons
- Trade promotions are effective for products whose holding cost at retailers is high and for which customer demand is sensitive to price



11.7. Managing Multi-Echelon Cycle Inventory

- Synchronization of the production for a manufacturer and the replenishment for a retailer lowers the cycle inventory in the supply chain (Cross-docking)
- Inventory profile at manufacturer and retailer w/o synchronization



$Q + Q/2$
VS
 $Q/2$



11.7. Managing Multi-Echelon Cycle Inventory

- **Synchronized integer replenishment policies** lower the cycle inventory in a multi-echelon supply chain
- Between distributor and retailer:
 - Integer multiples of **ordering frequency** across two stages facilitates a high level of cross-docking between them
 - Integer multiples of **lot sizes** across two stages allow for a portion of the delivery to be **cross-docked** between them



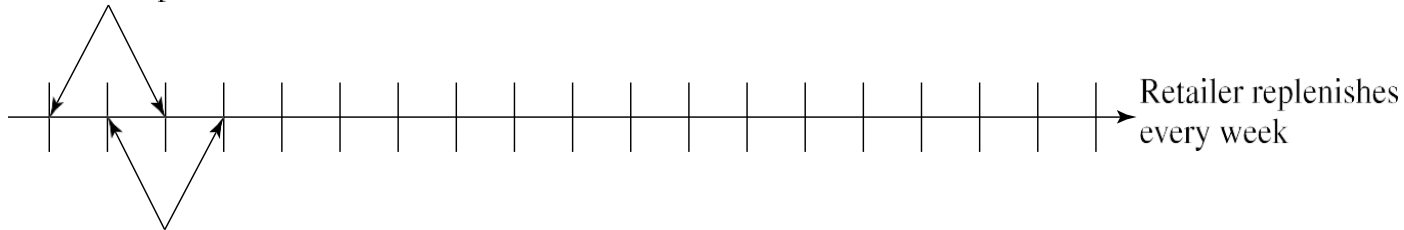
11.7. Managing Multi-Echelon Cycle Inventory

- Illustration of an Integer Replenishment Policy

Distributor replenishment order arrives



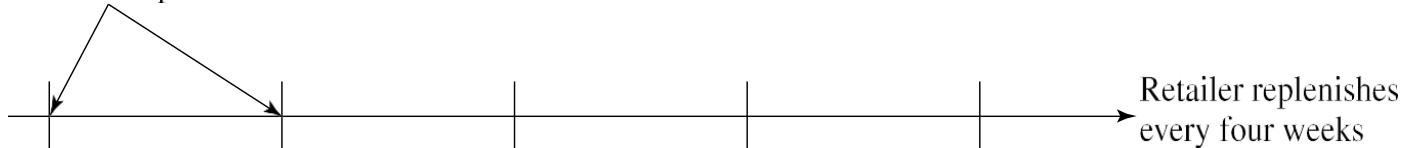
Retailer shipment is cross-docked



Retailer shipment is from inventory



Retailer shipment is cross-docked



11.8. Managing Levers to Reduce Cycle Inventory

- Levers to reduce cycle inventory (or lot size)
 - Reduce fixed ordering (including transportation) costs per order by aggregating fixed costs across multiple products, suppliers, or retailers
 - Implement volume-based discounting schemes rather than lot size-based discount schemes
 - Volume-based quantity discounts on rolling horizon
 - Discourage forward buying when offering promotion to retailers
 - EDLP, based on sell-through, manufacturer coupons
 - Implement vendor-managed inventory (VMI) policy
 - Implement a synchronized integer replenishment policy

